

Response to Reviewer A's comments

on paper “ A novel scaling indicator of early warning signals helps anticipate tropical cyclones ”

by *J. Prettyman, T. Kuna, and V. N. Livina*

We would like to thank the Reviewer for the positive opinion about the novel indicator. We have revised the paper according to the reviewer's comments as follows.

Comment 1. *The paper follows ideas proposed by Scheffer et al. (ref. 6), which uses the well-known fact that in the special case of a system represented by a variable $x(t)$ approaching a saddle-node bifurcation, the auto-correlation of $x(t)$ will increase prior to the bifurcation (critical slow down) as $t \rightarrow t_c$. [...] Obviously, since the power spectrum is the Fourier transform of the auto correlation and, in case of a scaling signal, the scaling exponents are directly related to the DFA exponent (eq. 4) this can be done. [...] For this to be useful the relevant issue is to establish under which circumstances the proposed PS indicator is better or worse than the other EWS indicators. This discussion is completely lacking.*

Response. The aim of this paper is not to investigate analytically in which dynamical systems the PS-indicator would be better or worse than other EWS methods, but rather to show that, in principle, the PS-indicator may provide an EWS. This is done by comparison to the ACF(1) and DFA indicators - not to show that it is better or worse, but simply to show that it behaves similarly, which in theory it should (as is pointed out by the reviewer, above). A comment to this effect has been added to the introduction.

We would like to note that the present manuscript follows the ideas of the paper [Livina and Lenton, GRL 2007] rather than the later paper mentioned by the reviewer [Scheffer et al, 2009].

Comment 2. *On the case(s) where we have a theoretical foundation for how the EWS indicators behave (simple bifurcations), the question of length and resolution of the time series at hand in relation to estimating the different measures should be discussed. From figure 3 it is apparent that PS is useless as EWS for the pitchfork bifurcation, DFA is marginal and only ACF(1) can be used. This is because in order to avoid false positives at the 1sigma level (very low, should be at least 2sigma), the mean should be outside the 1sigma level (gray band in fig 3). That does not happen for PS before $t=600$.*

Response. The reviewer is right that length and resolution of the series are important aspects of time series analysis.

In the case of high-frequency sampling of data, it is likely that the autocorrelation would be fluctuating at very high level (close to maximal value 1) and not providing any meaningful EWS information. In such cases, for applying any EWS indicator based on scaling properties (ACF, DFA or PS), it would be useful to pre-process the data with some aggregation or averaging, similarly to [?]. However, this issue is not specific to the new PS-indicator. Moreover, in many cases a time series is pre-processed before calculating EWS indicators (for instance, using Gaussian filter or moving average), which immediately changes the absolute value of autocorrelation and other scaling exponents. This means that the critical value of autocorrelation changes, and it is no longer bounded by the critical value 1. In such cases, the increasing trend of an EWS indicator is more informative than its values, which is a shortcoming of the approach with such data pre-processing. So far, its impact on the EWS indicators has not been studied, either in the context of analytical rigour or in the context of proximity to a tipping point; this must be a topic of further research in the area of early warning signals.

On the other hand, the length of the series (always finite and often imposing finite-size-effect challenges) may be problematic in case of estimation, for instance, variance in fixed-size windows. As was demonstrated in Fig.2 (middle panel) of [?], using fixed-size window in studying a dataset with increasing scaling exponent may generate a decreasing variance indicator.

We have added this text in the Discussion section.

We have included a short discussion about the PS-indicator applied to the pitchfork bifurcation (figure 3), drawing attention to the fact that the mean does not rise above the 1sigma level.

Comment 3. *The simulation (eq. 6) is fine, and I think the right way of testing. It is identical to what we presented in (Ditlevsen and Johnsen, GRL 2010) in the generic saddle node bifurcation case. In this connection the authors should argue why they omit using the increased variance, which we demonstrate is the strongest EWS indicator. The authors may want to look at: (Peter D. Ditlevsen, “Tipping Points in the Climate System”, to appear in “Nonlinear and Stochastic Climate Dynamics” (Ed. Christian Franzke and Terence O’Kane), Cambridge Univ. Press, 2017).*

Response. We thank the reviewer for the reference to (Peter D. Ditlevsen, “Tipping Points in the Climate System”, 2017), and we have added it to our bibliography.

We considered only ACF(1) (as a proxy for the auto correlation scaling exponent) and DFA, besides the novel power spectrum (PS) indicator because the inspiration came from [Kantelhardt et al. 2001], which provides a relationship between these three scaling exponents, and because the paper was considered a

natural progression from [Livina and Lenton, GRL, 2007] where the DFA propagator was introduced. As was mentioned in the response to Comment 2, in some cases of changing scaling properties, the variance indicator may decrease prior to a bifurcation, as was demonstrated on artificial data studied with a fixed-size window.

Comment 4. *Beside the few generic cases of bifurcations, for other tipping points or critical transitions the statistical quantities behave differently. The idea of change in long-range correlations comes from phase transitions in statistical mechanics. Here the transitions are usually characterized by emergence of long range correlations (power law behaviour), from the disordered state with short range (exponential) correlations and not by change of a scaling exponent. Actually the main characteristic of these critical transitions is the uniqueness (universality) of the scaling exponents. This should be discussed much more rigorously, especially also when a geophysical fluid dynamical situation as passing of cyclones is analysed. What type of dynamics is this, and what kind of EWS should we expect?*

Response. The pitchfork bifurcation is chosen as an example of a system where we will definitely expect the ACF(1) indicator to provide a very good EWS, therefore allowing us to see the new indicator applied to a familiar system. The approaching cyclone is chosen as an example of a system which has not previously been studied using these tipping-point EWS methods, and for which the dynamics of the system are unknown, thus providing a "blind" experiment. The passing cyclone is not modelled as a phase transition, and it is not known whether there exists a critical point such that we might claim universality of the scaling exponents. Rather, we have a time series which superficially resembles other geophysical tipping-point examples (such as in [Dakos et al. *Slowing down as an early warning signal for abrupt climate change*, PNAS, 2008]), thus we wish to experiment with our existing EWS methods, and then compare the performance of our new indicator. In this sense, it is an heuristic investigation of the properties of the new technique, with comparison with similar methods. We report our findings to facilitate further investigation of this technique.

We have added this text into the manuscript.

Comment 5. *What does "scaling properties similar to ACF" mean? In the standard cases (including eq. 6) the ACF is exponential, and the PS has two scaling regimes $\beta=0$ and $\beta=2$. This means that the PS indicator related to critical slow down is β increasing from -2 to 0.*

Response. In the phrase "Since DFA provides a method to study the scaling properties, similarly to the ACF, the DFA scaling indicator has been introduced...", we are

making the point that the DFA indicator provides a method to study longer-range scaling properties when they can be estimated, and in this respect it is similar to ACF. This is applicable to the changing scaling during the critical slow-down, when ACF ceases to decay exponentially and demonstrates scaling properties, similarly to DFA and PS. To clarify this, we added a note on slowing down in the same sentence.

Comment 6. *Abstract: Define “critical mode”. 2 lines after eq. 1: “uncorrelated” should be “short range correlated”.*

Response. We have changed the manuscript to accommodate the suggestions (critical modes, short-range correlated).

Comment 7. *How are data in Fig 1 generated? (I guess it is similar to Fig 2).*

Response. The data in fig.1 are indeed generated similarly to fig.2, we have added a note on this in the revised manuscript.

Comment 8. *Eq.5: I would have preferred something like a fractional Brownian motion (fBm) or some other less artificial standard process (and a discussion of why the usual Ornstein-Uhlenbeck red noise process is not relevant).*

Response. We have chosen this artificial time series, rather than the Ornstein-Uhlenbeck process, to reflect the result in [Livina et al 2012] and to provide an example specifically suited to the PS-indicator. A note of this has been included in the manuscript.

Comment 9. *Eq. 6: It should be discussed why the pitchfork bifurcation is studied rather than the generic saddle node. Pitchfork is special: It requires a perfect symmetry (and turns into saddle node with any small symmetry breaking perturbation). In perfectly symmetric systems, we often know more about the dynamics such that bifurcations can be predicted more precisely than what can be done with EWS.*

Response. The pitchfork bifurcation was chosen because in this case ACF(1) provides a clear EWS. Therefore, it is a good testbed for the PS-indicator, by analogy.

We have included here a similar experiment using a saddle node bifurcation described by the stochastic differential equation

$$\dot{z}(t) = -\frac{\partial}{\partial z} \left(\frac{z^4}{4} - \frac{z^2}{2} - (0.003t - 2)z \right) + \sigma\eta_t, \quad (1)$$

where $\sigma = 0.2$ and the η_t are independent samples from a Gaussian distribution. Equation 1 is integrated numerically with a time step of $\delta t = 0.05$ from $t = 1$ to $t = 1000$, and sampled with $\Delta t = 1$. In the range $t < 795$ the system has only

one stable node in the region $z < 0$, at $t \approx 540$ the system becomes a double well with another stable node in the region $z > 0$, but the noise is small enough that the system does not jump between stable nodes. The system then bifurcates after at $t \approx 795$ when the first stable node disappears, forcing the system to roll into the new stable state.

The system is integrated 50 times to produce 50 time series of length 1000, see fig. 1a. The EWS indicators are then applied to the 50 time series and the mean of all 50 indicator series is plotted with error bars of one standard deviation. We have performed this with the ACF(1)- and DFA-indicators (fig. 1b,c) and the new PS-indicator (fig. 1d). The ACF(1)-indicator provides a clear EWS as expected, there is a definite increasing trend starting at around $t = 750$.

We see that the result is similar to figure 3 in the manuscript, with ACF(1) giving the clearest EWS, and all three indicators rising (to some extent) before the event time $t = 795$, and the PS-indicator has a very large variance. However, it appears that the rise in all indicators can be explained by the fact that the system often jumps to the other stable node before the genuine bifurcation occurs. With $\sigma > 0.2$ this effect is obviously greater. With smaller values of σ we can be sure that the jump is not purely noise driven, since for very small σ the jump does not occur until the exact moment of the bifurcation, but in this case it is clearly very difficult to detect any warning signal. With the pitchfork bifurcation we avoid this problem of distinguishing bifurcation-driven and noise-driven jumps because the “other” stable state does not exist until the bifurcation. We therefore prefer it for this specific task of comparing indicator performance (without a view to claiming that any is better or worse than another).

We hope that the revised manuscript is now suitable for publication in EPL.

Yours sincerely,

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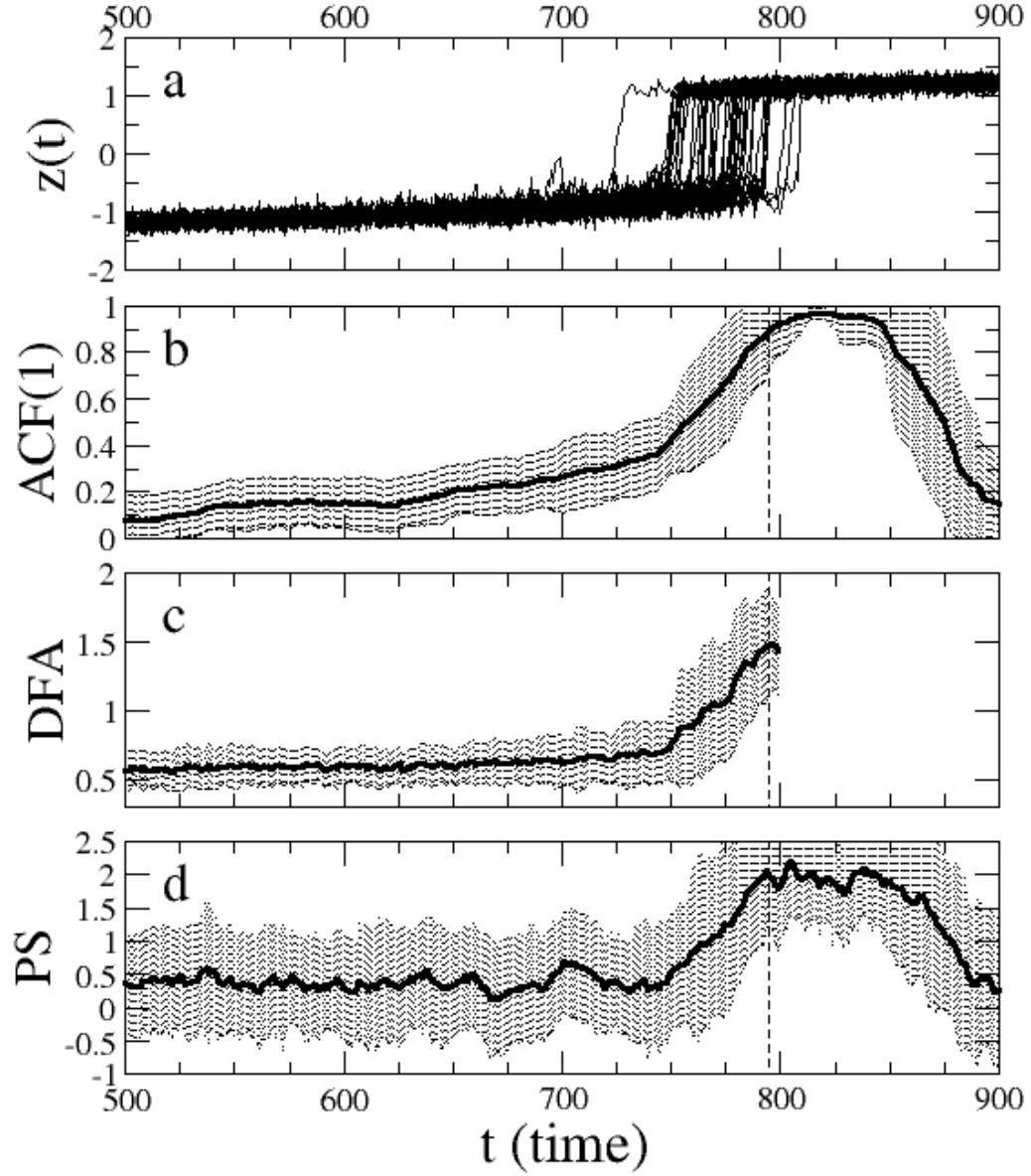


Figure 1: ACF(1)-, DFA- and PS-indicators applied to model data. (a) Data from 50 runs of the saddle node bifurcation model (see equation 1). (b,c,d) The mean ACF(1), DFA and PS-indicators (window size 100), with error bars of one standard deviation. The values of all the indicators begin to rise before the bifurcation, as expected, at around $t = 750$, although it is most noticeable for the ACF(1)-indicator. Note also that the variance is much larger in the PS-indicator.